

HINDAV DESHMUKH

(Machine Learning Engineer)

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SUMMARY

Machine Learning Engineer with 1.5+ years of experience building predictive models and deploying end-to-end ML systems. Skilled in Python, SQL, Scikit-Learn, TensorFlow, and NLP with hands-on experience in healthcare analytics and AI-driven applications. Experienced in data preprocessing, feature engineering, model training, and deploying ML APIs using Flask and FastAPI. Passionate about building scalable AI systems and production-ready ML pipelines.

ACHIEVEMENTS

- Achieved 92% accuracy in an ADR prediction system using ensemble ML models.
 - Improved customer support automation using RAG pipelines and semantic vector search.
 - Developed and deployed AI/ML and GenAI applications for real-world use cases.
 - Published a Deep Learning research paper on Movie Character Recognition in IJARSCT.
 - Built scalable ML and RAG systems using FastAPI, TensorFlow, LangChain, and Streamlit.
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SKILLS

Languages: Python, SQL

Machine Learning & Deep Learning: Scikit-Learn, XGBoost, TensorFlow, Keras, Regression, Classification, Feature Engineering, Feature Selection, Model Training, Model Evaluation, Hyperparameter Tuning, Ensemble Learning, CNNs, LSTMs, Transformers.

Generative AI & NLP: RAG Pipelines, LangChain, FAISS, Vector Databases, Vector Search, Hugging Face, OpenAI APIs, LLM Applications, Prompt Engineering, spaCy, NLTK

Backend & Deployment: FastAPI, Flask, REST APIs, Model Deployment, Streamlit

Databases & Cloud: MySQL, MongoDB, AWS EC2, AWS IAM, Linux

Tools & Visualization: Git, Jupyter Notebook, Google Colab, VS Code, Pandas, NumPy, Matplotlib, Plotly

EXPERIENCE

Newgen Technomate LLP – Pune

Machine Learning Engineer | Sep 2024 – Feb 2026

Adverse Drug Reaction (ADR) Prediction System

- Developed an Adverse Drug Reaction (ADR) prediction system using **Random Forest** and **Gradient Boosting** models, achieving **92% prediction accuracy** across healthcare datasets.
- Built scalable preprocessing and feature engineering pipelines using Pandas and NumPy for clinical and pharmaceutical data.
- Implemented model training, validation, and hyperparameter tuning workflows to improve prediction reliability and performance.
- Performed **exploratory data analysis** (EDA), feature selection, and correlation analysis to reduce overfitting.
- Deployed ML inference services using Flask and FastAPI APIs, enabling real-time healthcare risk prediction capabilities.

AI Customer Support Intelligence System

- Developed NLP pipelines for automated customer ticket classification and priority prediction using **Transformer-based** models.
 - Built **Retrieval-Augmented Generation** (RAG) workflows enabling contextual chatbot responses from internal knowledge base documents.
 - Designed scalable FastAPI inference services handling real-time customer query processing and analytics.
 - Implemented semantic vector search using **FAISS** for contextual retrieval across support documents and FAQs.
 - Created monitoring dashboards for ticket trends, customer sentiment, and SLA analytics using Streamlit and Plotly.
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KEY RESPONSIBILITIES

- Data collection and preprocessing of healthcare datasets.
 - Integrated and deployed model via web application.
 - Supported AI-based ADR risk prediction system.
 - Model evaluation and hyperparameter tuning.
 - Conducted EDA and feature engineering.
 - Developed ML models (Random Forest, Gradient Boosting)
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PROJECTS

StoxAi – AI/ML Stock Prediction System

Scope: ML-powered stock forecasting and sentiment analytics platform for real-time market insights.

- Developed a hybrid stock prediction system using LSTM models and GPT-4 sentiment analysis, achieving 72% forecasting accuracy across 200+ market records.
- Built scalable ETL and preprocessing pipelines for historical stock data, technical indicators, and real-time sentiment feeds.
- Designed and deployed FastAPI-based ML inference services handling 1k+ prediction requests/day with low-latency responses.
- Created interactive dashboards using Streamlit and Plotly for live analytics, trend visualization, and AI-driven investment insights.
- Tech Stack: TensorFlow, FastAPI, OpenAI API, Streamlit, Plotly, Python.

AskMyDocs – GenAI Document Conversationalist

Scope: GenAI-powered RAG platform for conversational querying and contextual analysis of PDF documents.

- Developed a Retrieval-Augmented Generation (RAG) system enabling conversational interaction with 100+ PDF documents using vector-based semantic search.
 - Implemented document chunking, embeddings, and FAISS vector indexing to improve contextual retrieval accuracy and response relevance.
 - Integrated LLM-powered query pipelines using LangChain for intelligent document summarization, Q&A, and contextual responses. Built an interactive Streamlit interface supporting real-time document upload, chat history, and multi-document querying.
 - Tech Stack: LangChain, LLMs, FAISS, Streamlit, Python, RAG.
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EDUCATION

- B.E. in Computer Science and Engineering | Jawaharlal Darda Institute of Engineering and Technology (2020 – 2024)
CGPA: 7.23
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